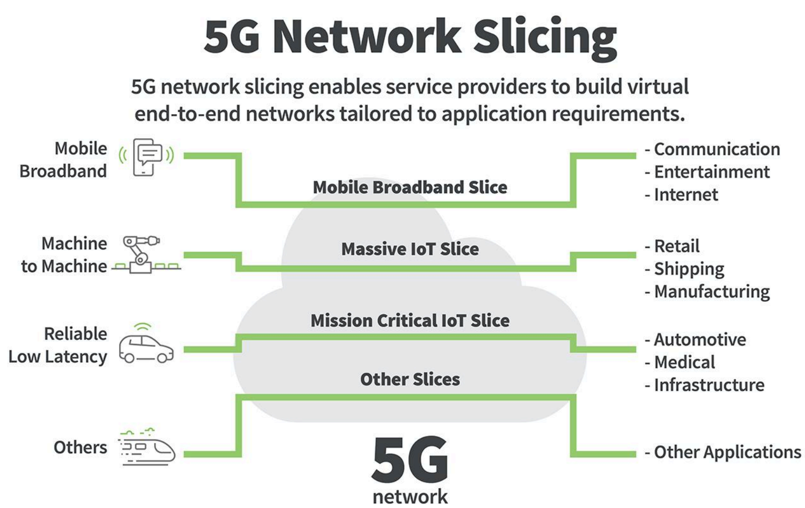
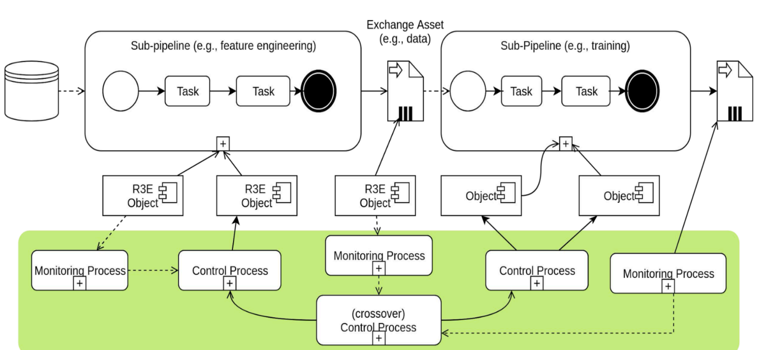
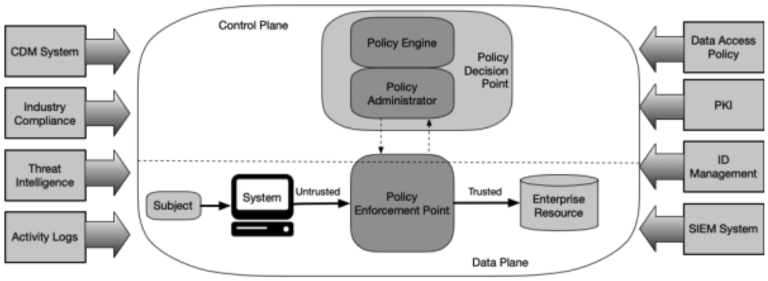
R3E in terms of a multi-continuum environment is in the core mission of modern telecom networks as they transition to a 5G (and even 6G) environment. For a while, 5G networks were not working standalone as they had to account for previous 4g LTE environments and base stations. This creates a very interesting dynamic where there is this existing 4G infrastructure and on top of it is 5g integration with more and more IoT edge devices seeking to integrate upon it. In this environment where there can be 100000 devices in a single factory or a city hub, R3E is a vital requirement, one can say it's the most important environment for any service provider to maintain trust and customers. As one can see from the diagram for 5g network slicing, ensuring that every single service remains elastic, robust, reliable and resilient is simply not an option but a mandatory requirement. This is not before we mention the rise of AI/ML integration that's been adopted across all industries. In the paper MLSys: The New Frontier in Machine Learning Systems, the three bottlenecks of full stack ML are listed as deployment, cost and accessibility, all of which telecom providers must handle with 5G to successfully create AI-native designs in preparation for future standards (6G is stated to be fully AI-native, which stresses the R3E principles even further)



One of the major things within the structure of R3E that relates to my major/interests is the resilience of AI/ML systems.



- R3EObject: encapsulate (sub)systems that can be controlled
- Control processes: can be complex with AI/ML-based algorithms



Considering these 2 diagrams, one taken from the slides about utilizing control processes to maintain R3E and NIST's official standards for a Zero-trust security unit. There are similarities between the two from my understanding. Both involve a monitoring/decision process (Policy Decision Point) and a control process (Policy Enforcement Point) that ensures the system operates according to the desired specifications without the risks of malicious actors seeking to destabilize it. These actors can be even the computing devices themselves. In the paper On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents, the author introduces methods that can turn agents into malicious actors through the use of clever prompt injections. This is especially dangerous in terms of security architecture as dangerous signals can now be flagged as safe. On the other hand, steps to ensure resiliency in ML systems are not well-defined (NIST Zero-trust standard is recently released in 2023) and proposals remain in theoretical research. It will be interesting to see how the literature develops in the future. We might be able to see what Nvidia envisions in terms of a purely agentic network where agents are doing the attacking and defending without human interference across a massive multi-continuum network.